Hanwen Chen

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EDUCATION

University of California, Santa Cruz *Expected Dec 2026*
M.S. in Electrical and Computer Engineering *Santa Cruz, CA*

Shanghai University *Jun 2025*
B.E. in Electronic Information Engineering *Shanghai, China*

EXPERIENCE

Tactile Manipulation Lab, UC Santa Cruz *Sep 2025 – Present*
Graduate Researcher | Autonomous Tomato Seedling Transplantation *Santa Cruz, CA*

- Building a seedling transplantation system on a **Franka Research 3** with a fixed ZED 2i and rubber-padded gripper; demonstrated stem-based acquisition, tray extraction, transport, and placement in initial end-to-end trials.
- Annotated main-stem instance masks on **200+ RGB-D images** with plant/session-level data splits to prevent leakage; fine-tuned an **Ultralytics YOLO** segmentation model and skeletonized masks (OpenCV, scikit-image) into centerlines to select long, low-curvature grasp segments.
- Recovered 3D stem position and tangent from ZED SDK point clouds, filtering low-confidence and depth-discontinuity pixels before **RANSAC + PCA** line fitting (NumPy/SciPy); debugged geometry in Open3D.
- Calibrated camera-to-base and tool-to-contact-center transforms and validated them on held-out targets; generated side-approach grasp candidates perpendicular to the stem tangent, filtered by workspace, **inverse kinematics**, and gripper–tray collision clearance, and ranked by depth quality and capture tolerance.
- Implemented a Python **state machine** (observe, validate, pre-grasp, acquire, extract, transport, place, recover) over **franky/libfranka** with bounded velocity, acceleration, and jerk; used external-wrench estimates to detect unexpected contact, plus re-observation, bounded retries, and retreat.
- Measured gripper capture tolerance via controlled 3-axis offsets, testing holding separately from tray extraction to isolate slippage vs. extraction resistance; used results to set localization requirements.

Zentek *Jun 2024 – Dec 2024*
Systems Engineering Intern *Shanghai, China*

IMU-Driven Digital Twin GitHub

- Built a **100 Hz** IMU pipeline on Jetson Orin Nano with an MPU6050 over I2C; estimated roll, pitch, and yaw with Kalman filtering and removed gravity from acceleration, keeping measured orientation error within **2°**.
- Streamed sensor data to the host over HTTP and converted it into OpenUSD transforms published through Omniverse Nucleus Live collaboration sessions, achieving **sub-100 ms** measured scene latency.

Real-Time Conversational Digital Human GitHub

- Built a FastAPI/WebSocket pipeline linking Riva ASR, a streaming LLM, GPT-SoVITS, and Audio2Face (gRPC) to a React/Three.js avatar; pipelined sentence-level speech and animation so playback began before the full response was ready, cutting perceived latency by **60%**.
- Ran TTS and facial-animation inference in producer–consumer worker threads alongside asyncio I/O; added Audio2Face audio chunking for buffer limits, an STFT-based procedural fallback, and WebSocket auto-reconnect.

TECHNICAL SKILLS

Languages & software: Python, C++, JavaScript, Bash, Linux, Git, FastAPI, asyncio, WebSockets, gRPC
Robotics: Franka Research 3, franky/libfranka, camera-to-robot calibration, rigid-body transforms, inverse kinematics, grasp planning, state machines, Kalman filtering, Jetson Orin
Perception: PyTorch, Ultralytics YOLO, OpenCV, scikit-image, ZED SDK, Open3D, NumPy/SciPy, RANSAC/PCA
Simulation & hardware: NVIDIA Omniverse Kit, OpenUSD, Nucleus Live, Onshape, SolidWorks, 3D printing